Conveyor Maintenance and Jam Troubleshooting Guide

Overview

This guide covers routine maintenance and jam troubleshooting for conveyors C1 through C4 at Riverbend Packaging. The conveyors are chain-driven table-top lines with variable-frequency drives, guide rails, and transfer plates between sections. Keeping the lines clean and properly tensioned prevents most jams.

Specifications

Item	C1-C2	C3	C4
Chain type	Plastic table-top	Plastic table-top	Roller chain
Drive	1 hp VFD	1 hp VFD	2 hp VFD
Normal speed	60 ft/min	60 ft/min	40 ft/min
Jog speed	8 ft/min	8 ft/min	6 ft/min

Preventive Maintenance

Interval	Task
Daily	Remove debris and glass; check guide rail alignment
Weekly	Lubricate chain with approved dry lube; check tracking
Monthly	Inspect sprockets, wear strips, and idlers
Quarterly	Check chain tension and drive coupling
Annually	Replace worn chain sections; inspect gearbox oil

Chain Tension

Correct tension allows the chain to lift about 0.5 inch at mid-span. Loose chain jumps sprockets and causes stalls; tight chain overloads bearings.

1. Lock out the conveyor.
2. Loosen the take-up bolts.
3. Adjust until sag is correct and equal on both sides.
4. Tighten bolts and rotate by hand to confirm free movement.

Jam Troubleshooting

Jam location	Common cause	Fix
Filler infeed	Bottles tipped by worn guide rail	Realign or replace rail
Transfer plate	Gap too large or plate worn	Adjust gap to 1/8 inch
Capper infeed	Cap or bottle debris	Clear and clean chute

Jam location	Common cause	Fix
Outfeed to packer	Packer stall backs up line	Check packer first
Return loop	Broken glass or labels	Clean return belt
Mid-line stall	Seized idler or overload	Replace idler; check amp draw

Clearing Small Jams

Small jams, such as a single fallen bottle on a straight section, can be cleared quickly with the conveyor in jog mode at reduced speed. Use the jog button on the local station to move the chain slowly and use a jam hook to free the container. Release the jog button as soon as the bottle is free. This keeps downtime short and avoids a full line restart.

For larger jams involving several bottles, a jammed chain, or a jam at a transfer point, stop the conveyor and use the isolation points at the disconnect before reaching in.

Diagnosing Repeated Jams

1. Note the jam location and time on the downtime log.
2. Check guide rails for rubbing marks on bottles.
3. Measure the transfer plate gap with feeler gauges.
4. Confirm the line speeds between sections are matched within 2 percent.
5. Inspect the chain for stretched links.

Motor and Drive Issues

Symptom	Likely cause	Corrective action
VFD overcurrent fault	Jam or tight chain	Clear jam; check tension
Conveyor slow	Incorrect VFD parameter	Reset speed parameter
Motor overheating	Blocked fan or overload	Clean fan; check load
Gearbox noise	Low oil or worn gears	Check oil; schedule rebuild

Replacing a Chain Section

1. Lock out the conveyor and release tension.
2. Locate the damaged section and remove the pin.
3. Install a new section of the same pitch and material.
4. Reinsert the pin and retaining clip.
5. Restore tension and run empty for ten minutes.

Wear Parts

Keep stocked: wear strips, plastic chain sections, guide rail brackets, idler bearings, and dry lube. Replace wear strips when worn to 1/16 inch thickness.

Records

Log all jam events with location, duration, and cause. Review weekly for trends. Repeated jams at one location trigger a maintenance work order.